EMC ASSESSMENT (UKCA – RADIO EQUIPMENT)

Product: SidecarTridge Croissant Keyboard Emulator

Model/Revision: SidecarTridge Croissant Keyboard Emulator for Atari ST/STe / REV1.1.0

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1. PRODUCT OVERVIEW

The product is a low-voltage digital electronic device intended for use in retro computing systems. It is designed as an internal or semi-internal component (e.g. expansion board, cartridge, or interface module) and is powered by the host system.

The product integrates a radio module providing wireless functionality (e.g. Bluetooth and/or Wi-Fi) in the 2.4 GHz band.

Typical operating voltage is below 50 V DC (e.g. 3.3V / 5V / 12V).

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2. APPLICABLE REGULATIONS

This assessment is made in relation to:

- Radio Equipment Regulations 2017 (S.I. 2017/1206)

The Electrical Equipment (Safety) Regulations 2016 do not apply separately, as their essential requirements are covered within the Radio Equipment Regulations.

The Electromagnetic Compatibility Regulations 2016 do not apply separately, as EMC requirements are included within the Radio Equipment Regulations.

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3. DESIGN CHARACTERISTICS RELEVANT TO EMC AND RADIO PERFORMANCE

- The device contains standard digital logic components (e.g. CMOS/TTL, microcontrollers, memory devices).
- The product integrates a radio module operating in the 2.4 GHz ISM band.
- The radio functionality (e.g. Bluetooth and/or Wi-Fi) is provided entirely by the integrated module.
- No modifications affecting radio frequency performance (antenna, RF path, shielding, or firmware parameters) have been made to the module.
- The device does not incorporate high-frequency switching power supplies (unless provided externally by the host system).
- Signal frequencies of non-radio circuitry are limited to those typical of embedded or legacy computing systems.
- External connections are limited in length and are typically internal to the host system or via short cables.

- The device includes a microcontroller-based system operating at moderate clock frequencies typical of embedded systems.
- A USB interface is present for local data transfer; it is not intended for continuous high-speed communication during normal operation.

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4. RADIO MODULE INTEGRATION AND EMC ASSESSMENT

The product integrates a pre-assessed radio module providing wireless communication functionality.

Compliance with the essential requirements of the Radio Equipment Regulations 2017, including:

- Regulation 6(2) – Health and safety
- Regulation 6(3) – Electromagnetic compatibility
- Regulation 6(4) – Efficient use of radio spectrum

is ensured through the use of this module in accordance with the module manufacturer's integration guidelines.

The EMC performance of the radio functionality is supported by designated standards such as:

- BS EN 301 489-1 (EMC for radio equipment – general)
- BS EN 301 489-17 (EMC for broadband data transmission systems)

The host product:

- does not introduce additional continuous high-frequency emission sources
- does not alter the RF characteristics of the integrated module
- operates as a low-power internal digital system

- The USB interface is used intermittently for file transfer and does not represent a continuous high-frequency emission source.

Conclusion:
The product does not adversely affect the EMC performance of the integrated radio module when used as intended.

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5. RF EXPOSURE

Compliance with RF exposure requirements is ensured through:

- BS EN 62479 (assessment of low-power equipment)

The product operates at power levels consistent with the integrated radio module and is intended for use within enclosed host systems.

Conclusion:
The product meets the applicable requirements for human exposure to electromagnetic fields.

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6. INSTALLATION AND USE CONDITIONS

- The product is intended for integration into a larger electronic system.
- It is not designed to operate as a standalone end-user device.
- Proper EMC and radio performance depend on correct installation and use in accordance with the module manufacturer's integration guidelines.

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7. OVERALL CONCLUSION

Based on the use of a pre-assessed radio module and the design characteristics of the product, the device is considered to comply with the essential requirements of the Radio Equipment Regulations 2017 when installed and used as intended.

This assessment is supported by designated standards such as:

- BS EN 300 328 (radio spectrum use)
- BS EN 301 489-1 and BS EN 301 489-17 (EMC for radio equipment)
- BS EN 62479 (RF exposure)

No additional EMC mitigation measures are required based on the design and intended use.

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8. RESPONSIBLE PERSON

Name: Diego Parrilla Santamaria
Position: CEO
Company: GOODDATA LABS SL

Date: 23 April 2026

Signature: